



Arab Risk Monitor: assessing vulnerability and resilience in the region



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Economic and Social Commission for Western Asia

Arab Risk Monitor: assessing vulnerability and resilience in the region



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Key messages

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- *The overall level of conflict risk in the Arab region has deteriorated in the past decade.*
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- *Weak institutions and natural resource scarcity perpetuate exposure to instability and conflict in the Arab region.*
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- *Water scarcity, food insecurity and corruption were identified as the primary drivers of high vulnerability in the Arab region.*
-
- *In 2021, the main drivers of low resilience in the Arab region were limited social protection, water stress, and lack of voice and accountability.*
-
- *Risks are interconnected. Conflict does not occur in isolation.*
-
- *Vulnerability and resilience are highly correlated.*
-

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Introduction

The crises affecting the Arab region¹ are complex and protracted, exacerbated by ongoing conflicts and their widespread consequences on people, society and institutions. Structural economic changes and sudden shocks like COVID-19 have added other layers of challenges. During the past decade, the Arab region has witnessed high levels of destruction and loss of life, increasing inequalities and reducing public trust in the government to ensure participation in decision-making and delivery of basic services. This has led to a surge in humanitarian needs and food insecurity while economies are struggling with low productivity, macroeconomic instability and vulnerability to external shocks.

The issues caused by years of conflict are amplified by megatrends such as climate change, which disproportionately affects the most in need. All of this is happening in a global landscape of interconnected vulnerabilities and eroding resilience which has led to a polycrisis whereby multiple risks interact such that the overall impact far exceeds the sum of each one.

In this context, prevention requires new modes of long-term, multi-stakeholder, cross-pillar thinking that can address the multidimensional nature of risk. There is a growing need for approaches that take account of complex risks or are risk-informed.

Risk-informed policymaking in the Arab region is essential not only to prioritize and tackle current challenges but to foresee future risks and challenges, and design policies to mitigate them. To address the main drivers of risk that exacerbate development challenges in (conflict-affected) Arab countries through risk-informed policy, the United Nations Economic and Social Commission for Western Asia (ESCWA) developed the Arab Risk Monitor, which provides measures of risk grouped under three “risk pathways” that are associated with a greater risk of conflict, crisis and instability in the Arab region. Within each pathway, the Arab Risk Monitor report provides one or more risk domains:

- The Conflict Pathway looks at historical grievances as a predictor of future conflict as well as at a country’s enabling security environment. It produces one risk domain: Conflict Risk.
- The Climate Pathway looks at the impact of climate hazards as well as at the availability and management of natural resources. It produces two risk domains: Climate Hazard Risk and Natural Resource Risk.
- The Development Pathway represents the entire complex set of dynamics that in most of the cases are direct or indirect causes of risk in the Arab region. It produces three risk domains: Economic Risk, Social Risk and Institutional Risk.

¹ The countries included in the report are Algeria, Bahrain, Comoros, Djibouti, Egypt, Iraq, Kuwait, Lebanon, Libya, Mauritania, Morocco, Oman, State of Palestine, Qatar, Saudi Arabia, the Sudan, Somalia, Syrian Arab Republic, Tunisia, United Arab Emirates and Yemen. Fragile and conflict-affected situations as defined by the World Bank include Iraq, Lebanon, Libya, State of Palestine, Somalia, the Sudan, Syrian Arab Republic and Yemen. Gulf Cooperation Council countries include Bahrain, Kuwait, Oman, Qatar, Saudi Arabia and the United Arab Emirates.

Figure 1. Conceptual framework for the Arab Risk Monitor



Source: ESCWA elaboration.

The six risk domains are expressed as a combination of two dimensions: increasing vulnerability and decreasing resilience. ESCWA defines “vulnerability” in terms of structural exposure to shocks, and “resilience” in terms of policy-driven capacity to absorb the impact of shocks (figure 1).²

For each risk domain, an aggregate score is produced by ESCWA based on a composite model that measures vulnerability and resilience according to the matrix (table 1).³

Table 1. Levels of risk

		Level of Resilience				
		Very Low	Low	Medium	High	Very High
		0–0.20	0.21–0.40	0.41–0.60	0.61–0.80	0.81–1
Level of Vulnerability	Very High	Severe	Severe	Significant	Moderate	Moderate
	0.81–1					
	High	Severe	Significant	Significant	Moderate	Moderate
	0.61–0.80					
	Medium	Significant	Significant	Moderate	Minor	Minor
	0.41–0.60					
	Low	Moderate	Moderate	Minor	Minor	Negligible
	0.21–0.40					
	Very Low	Moderate	Moderate	Minor	Negligible	Negligible
	0–0.20					

Source: ESCWA elaboration.

2 For more details on the conceptual framework see ESCWA (2023) *Arab Risk Monitor: A Conceptual Framework*. Available at <https://www.unescwa.org/publications/arab-risk-monitor-conceptual-framework>.

3 For more details on the methodology see ESCWA (2023) *Arab Risk Monitor: Quantifying the drivers of risk of conflict, version 1.0*. Available at <https://www.unescwa.org/publications/arab-risk-monitor-drivers-conflict>.

This paper is part of a suite of knowledge products produced under the ESCWA initiative to support risk-informed policymaking. The report serves to enhance and coordinate the generation and sharing of evidence-based information and analysis for preventing and mitigating risks in the Arab region.

Arab Risk Monitor: A Conceptual Framework provides the theoretical framework that underpins the content and structure of this report. *Arab Risk Monitor: Quantifying the drivers of risk of conflict* offers a comprehensive methodology for assessing the risk of conflict by examining various drivers, which are categorized and analysed within the dimensions of vulnerability and resilience. The present report, *Arab Risk Monitor: assessing vulnerability and resilience in the region*, introduces the key results of the Risk Monitor and provides an overall analysis

of risk and its evolution in the Arab region since 2010.

The paper is mainly directed towards policymakers in the Arab region to integrate risk analysis into policymaking, programming and planning. The assessment also serves development and humanitarian entities such as United Nations agencies, other international organizations and non-governmental organizations in the spirit of the Sustainable Development Goals (SDGs) 16 and 17.

Section 1 of this report provides the key highlights. Section 2 presents the results related to the Conflict Pathway, section 3 focuses on the Climate Pathway and section 4 explores the results concerning the Development Pathway. Section 5 examines the evolution of the risk landscape in the region through a comparative assessment of vulnerability and resilience over the past decade.

1. Highlights

A. Key message 1: The overall level of conflict risk in the Arab region has deteriorated in the past decade

As of 2021, the Arab region is more vulnerable and less resilient to conflict than it was in 2010 (table 2). A key driver of deterioration is the Conflict Risk domain (+10 per cent vulnerability and -11 per cent resilience compared to 2010 levels). This degradation is mainly attributed to the eruption and persistence of armed conflicts in the last decade. The Arab region also saw a marked deterioration in the Economic Risk domain (+9 per cent vulnerability and -15 per cent

resilience compared to 2010 levels). Due to increased reliance on foreign trade and capital markets, along with a contracting fiscal space, Arab states are now more vulnerable and susceptible to the impact of shocks than they were in the previous decade.

The Climate Hazard Risk domain recorded a deterioration caused by the rising human impact of natural disasters. However, the region is increasingly building resilience to climate hazards, largely due to increasing climate adaptation strategies. Only the Social Risk domain improved in 2021 driven by expanding social safety nets and decreasing infant mortality rates.

Table 2. Evolution of risk in the Arab region

Pathway	Risk Domain	Dimension	2010	2015	2021	2021 vs. 2010	
Conflict	Conflict Risk	Vulnerability	0.29	0.41	0.32	10%	Deteriorated
		Resilience	0.48	0.44	0.43	-11%	Deteriorated
Climate	Climate Hazard Risk	Vulnerability	0.11	0.16	0.18	59%	Deteriorated
		Resilience	0.30	0.35	0.47	56%	Improved
	Natural Resource Risk	Vulnerability	0.47	0.49	0.49	5%	Deteriorated
		Resilience	0.39	0.39	0.38	-4%	Deteriorated
Development	Economic Risk	Vulnerability	0.39	0.40	0.43	9%	Deteriorated
		Resilience	0.65	0.59	0.55	-15%	Deteriorated
	Social Risk	Vulnerability	0.37	0.34	0.33	-9%	Improved
		Resilience	0.50	0.53	0.51	4%	Improved
	Institutional Risk	Vulnerability	0.66	0.69	0.70	7%	Deteriorated
		Resilience	0.43	0.41	0.40	-8%	Deteriorated

Source: ESCWA elaboration.

B. Key message 2: Weak institutions and natural resource scarcity perpetuate instability and conflict in the Arab region

The Arab region's primary risks for 2021 are the Natural Resource Risk and the Institutional Risk domains (table 3). The Institutional Risk domain is ranked as "significant" with a high level of vulnerability (0.70 in 2021) and a medium level of resilience (0.40 in 2021). Ineffective institutions can create a breeding ground for conflict. Corruption can erode trust in institutions and undermine the legitimacy of the state. Without effective mechanisms for participating in the decision-making process and

holding leaders accountable, people may resort to extreme measures, including violent conflict, to demand change.

At the same time, the Natural Resource Risk domain is ranked as "significant" with a medium level of vulnerability (0.49 in 2021) and a low level of resilience (0.38 in 2021). The Arab region relies heavily on agriculture which is highly vulnerable to climate change. Water scarcity is a pressing issue and a threat to the region's sustainability. The control of natural resources can ignite conflict between different actors. In areas where water is scarce, poor management practices, lack of conservation and inefficiencies can also create social tensions.

Table 3. Overview of risk in the Arab region

			2010	2015	2021
Conflict Risk	Minor	Vulnerability	0.29	0.41	0.32
		Resilience	0.48	0.44	0.43
Climate Hazard Risk	Minor	Vulnerability	0.11	0.16	0.18
		Resilience	0.30	0.35	0.47
Natural Resource Risk	Significant	Vulnerability	0.47	0.49	0.49
		Resilience	0.39	0.39	0.38
Economic Risk	Moderate	Vulnerability	0.39	0.40	0.43
		Resilience	0.65	0.59	0.55
Social Risk	Minor	Vulnerability	0.37	0.34	0.33
		Resilience	0.50	0.53	0.51
Institutional Risk	Significant	Vulnerability	0.66	0.69	0.70
		Resilience	0.43	0.41	0.40

Source: ESCWA elaboration.

C. Key message 3: Water scarcity, food insecurity and corruption were identified as the primary drivers of high vulnerability in the Arab region

Water availability is an ever-increasing challenge in the Arab region, posing an existential threat to one of the world's most water-scarce areas. Climate change is exacerbating the water crisis with irregular rainfall patterns leading to more intense droughts, floods and other extreme weather events. Control over freshwater resources can fuel tensions and conflict between groups and communities. Out of 22 Arab countries, 9 display a “very high” vulnerability score (above 0.80) for the Water Scarcity indicator.

Corruption is a significant threat to peace and stability. As public resources are diverted, this

can create social and political tensions, especially in the presence of marginalized groups. Corruption can erode trust in institutions and undermine the legitimacy of the state. Out of 22 Arab countries, 8 display a “very high” vulnerability score (above 0.80) for the Corruption indicator.

Lack of access to sufficient food or food security can lead to social outbreaks and increase the likelihood of conflict. Over-reliance on food resources traded on international markets can increase vulnerability to price shocks caused by environmental factors. Demonstrations, riots and conflict are more likely when international and domestic food prices rise. Among the 22 Arab countries, 7 display a “very high” vulnerability score (above 0.80) for the Food Insecurity indicator (table 4).

Table 4. Drivers of high vulnerability in the Arab region

Vulnerability Drivers	2010	2015	2021
Water Scarcity	0.75	0.78	0.79
Corruption	0.66	0.69	0.70
Food Insecurity	0.56	0.59	0.65
Political Instability	0.52	0.58	0.56
Unemployment	0.40	0.44	0.49
Proliferation of Small Arms	0.47	0.47	0.47
Inequality	0.46	0.43	0.42
Trade Dependence	0.40	0.40	0.40
Youth Bulge	0.51	0.42	0.36
Neighbouring Conflict	0.30	0.63	0.36
Financial Dependence	0.15	0.18	0.24
Reliance on Agriculture	0.17	0.19	0.18
Disaster Impact	0.11	0.16	0.18
Infant Mortality	0.19	0.17	0.15
Conflict Intensity	0.07	0.19	0.11
Conflict History	0.09	0.20	0.11

Source: ESCWA elaboration.

D. Key message 4: In 2021, the main drivers of low resilience in the Arab region were limited social protection, water stress, and lack of voice and accountability

Social protection is essential for building resilience and preventing conflict. Social safety nets provide vulnerable communities with the resources needed to cope with shocks, reducing tensions and promoting social cohesion. Among the Arab countries with available data, nine have a “very low” resilience score (below 0.20) for the Social Protection indicator.

As the demand for water increases, particularly in rapidly urbanizing areas and for agricultural purposes, the withdrawal of freshwater from rivers, lakes and groundwater sources can exceed the natural replenishment rate leading to

the depletion of these resources. Poor water management practices, lack of water conservation, and inefficient use and allocation can lead to social tensions or economic challenges in areas where water is scarce. The Water Stress indicator reveals that in 10 out of the 22 Arab countries, the level of resilience is “very low” (below 0.20).

Promoting voice and accountability is essential for building resilience and preventing conflict. By providing citizens with access to information and opportunities to participate in decision-making processes, and by holding their leadership accountable, institutions can enhance their legitimacy, foster trust and social cohesion, and prevent conflict. According to the Voice and Accountability indicator, 6 of the 22 Arab countries have a “very low” resilience (below 0.20) (table 5).

Table 5. Drivers of low resilience in the Arab region

Resilience Drivers	2010	2015	2021
Social Protection		0.36	0.26
Water Stress	0.26	0.26	0.26
Military Capacity	0.35	0.31	0.26
Voice and Accountability	0.29	0.30	0.29
Adaptation Finance	0.31	0.35	0.33
Forced Displacement	0.46	0.39	0.36
Government Effectiveness	0.41	0.39	0.38
Fiscal Space	0.62	0.53	0.39
Health Coverage	0.32	0.36	0.41
Rule of Law	0.45	0.43	0.42
Land Stress	0.52	0.52	0.50
Education	0.48	0.51	0.55
Economic Development	0.61	0.60	0.59
Economic Growth	0.72	0.64	0.66
Adaptation Strategies	0.29	0.36	0.70
Water and Sanitation Services	0.69	0.74	0.77
Territorial Integrity	0.81	0.76	0.79

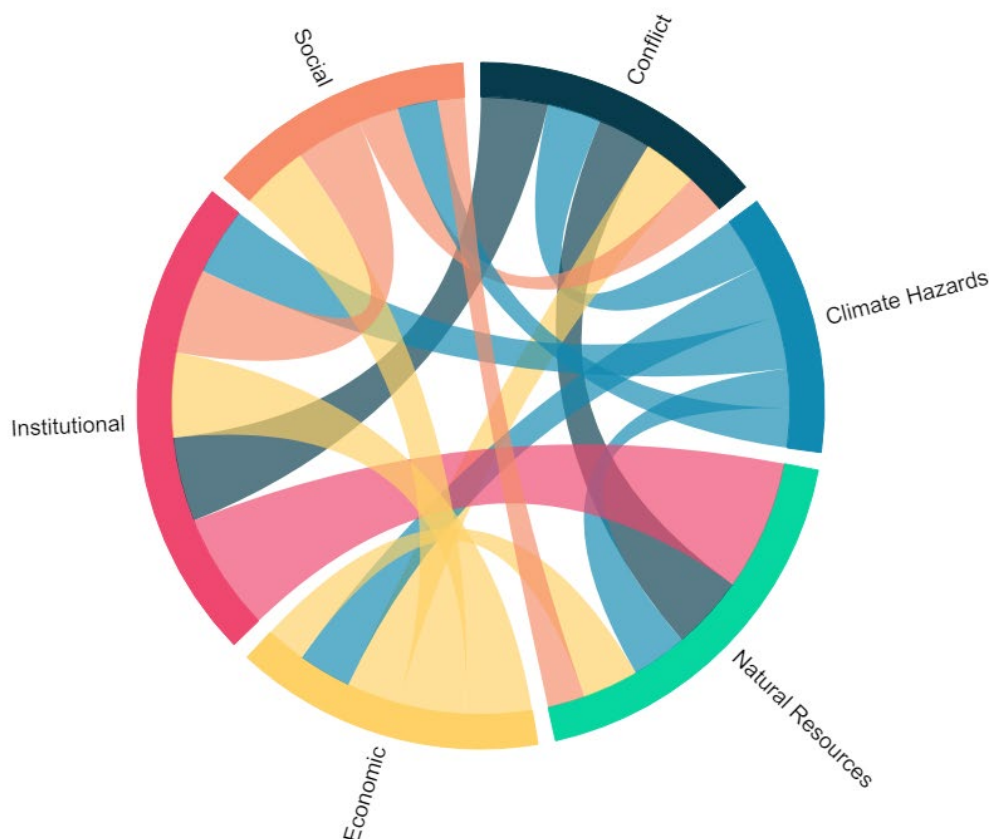
Source: ESCWA elaboration.

E. Key message 5: Risks are interconnected. Conflict does not occur in isolation

Conflict risk is intertwined with all other risk domains – climate hazards, natural resources, economy, society and institutions. Arab countries experiencing conflict are likely to suffer from the disintegration of their institutions in addition to grave economic, social and climate challenges. Figure 2 shows the interconnections among the six domains of risk. Each arch represents the number of

countries at a significant risk level or above for each domain. The thickness of each flow (or chord) represents the relative number of countries with significant risk for both connected domains. For example, all five countries with high levels of conflict risk also have high levels of institutional risk, whereas three out of the five countries also have high social risk levels. In other words, a country that exhibits a high level of risk in one domain is much more likely to exhibit high levels of risk in the other domains than a country with a lower level of risk.

Figure 2. Interconnectivity of risk in the Arab Region, 2021



Source: ESCWA elaboration.

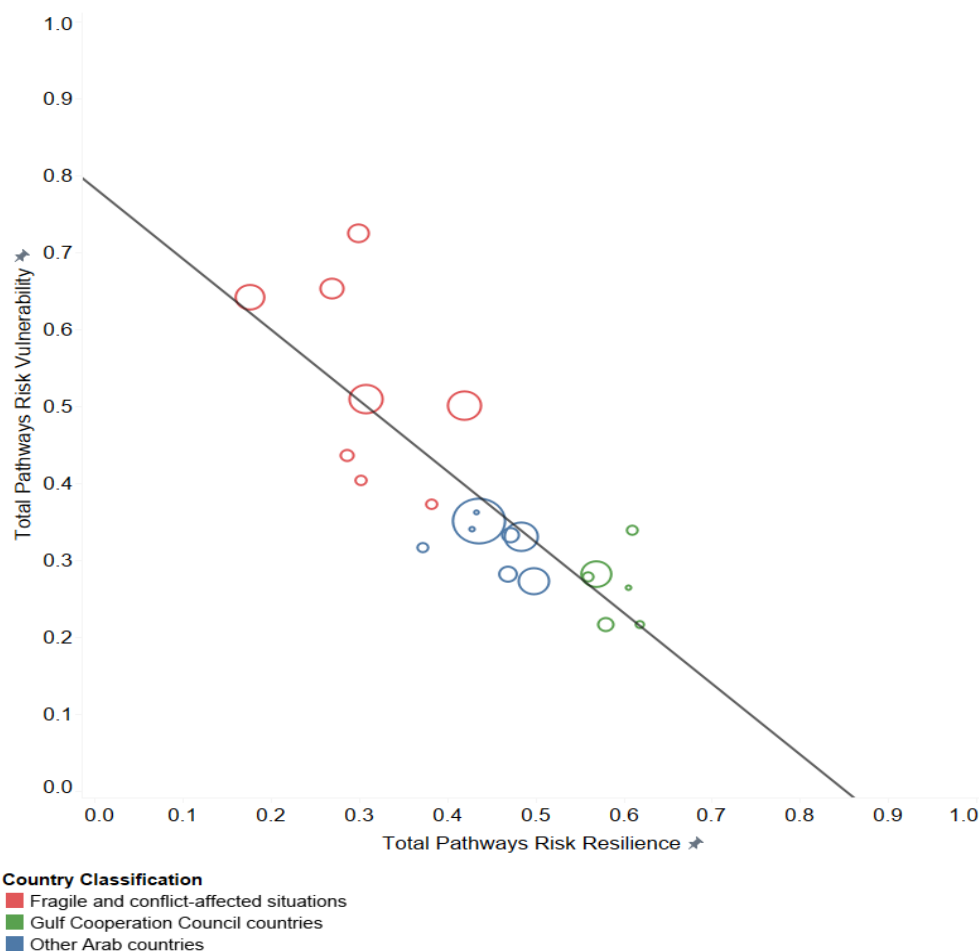
F. Key message 6: Vulnerability and resilience are highly correlated

As shown in figure 3, measures of overall vulnerability and resilience are correlated with a negative sloping trendline that shows countries with high vulnerability tend to also have low resilience to risks. Each circle represents an Arab country with the size of the circle denoting the population size relative to the region. The tight concentration or bunching of the circles around the trendline suggests that the correlation between vulnerability and resilience

is high. The values for both vulnerability and resilience are the average of the values for the three pathways: Conflict, Climate and Development.

In summary, conflict does not happen in a vacuum; different dimensions of risk are interconnected and are correlated. To address this, the Arab Risk Monitor provides a multidimensional and integrated framework designed to measure the different pathways and domains of risk and the interaction between the dimensions of risk.

Figure 3. Vulnerability and resilience in the Arab region, 2021



Source: ESCWA elaboration.

2. The Conflict Pathway

Conflict risk

Countries that have experienced conflict in the past are much more vulnerable to future conflict than those that have been at peace. Additionally, countries located in close proximity to areas of conflict are at higher risk of becoming embroiled in such situations due to factors such as porous borders, population displacement and social ties (table 6). On average, the Conflict Risk domain is ranked as "minor" with a low level of vulnerability (0.32 in 2021) and a medium level of resilience (0.43 in 2021). The relatively low regional score does not fully capture the persistence of substantial conflict-related challenges affecting multiple Arab countries. Conflict-related risk should not be underestimated. Notably, the regional score has deteriorated over the past decade, marked by a 10 per cent increase in vulnerability and an 11 per cent decrease in resilience compared to 2010 levels.

The lack of historical grievances can reduce the potential of internal conflict. In the Arab region, conflict-related vulnerability is the highest in countries facing conflict or protracted crises.

Due to prolonged fighting in the region, conflict-affected countries were more vulnerable in 2021 than they were in the previous decade. Conflict-related vulnerability has also increased in countries not currently affected by violent conflict. However, 9 out of 22 Arab countries had reduced their vulnerability since 2010.

The risk of conflict is not only influenced by a country's vulnerability but also its level of resilience. The movement of people fleeing conflicts and disasters can have significant effects on a country's capacity to manage future risks, particularly in densely populated urban areas. Limited government control over territory can also negatively impact a country's resilience. Conflict-related resilience is "very low" or "low" in seven Arab countries.

Governments can take steps to build resilience and mitigate conflict risk by involving citizens in decision-making processes and providing them with access to services and opportunities. These efforts can strengthen social cohesion and enhance a country's ability to withstand shocks.

Table 6. Conflict risk in the Arab region

Pathway	Risk Domain	Component	2010	2015	2021	Risk Level, 2021	2021 vs. 2010	
Conflict	Conflict Risk	Vulnerability	0.29	0.41	0.32	Minor	10%	Deteriorated
		Resilience	0.48	0.44	0.43		-11%	Deteriorated

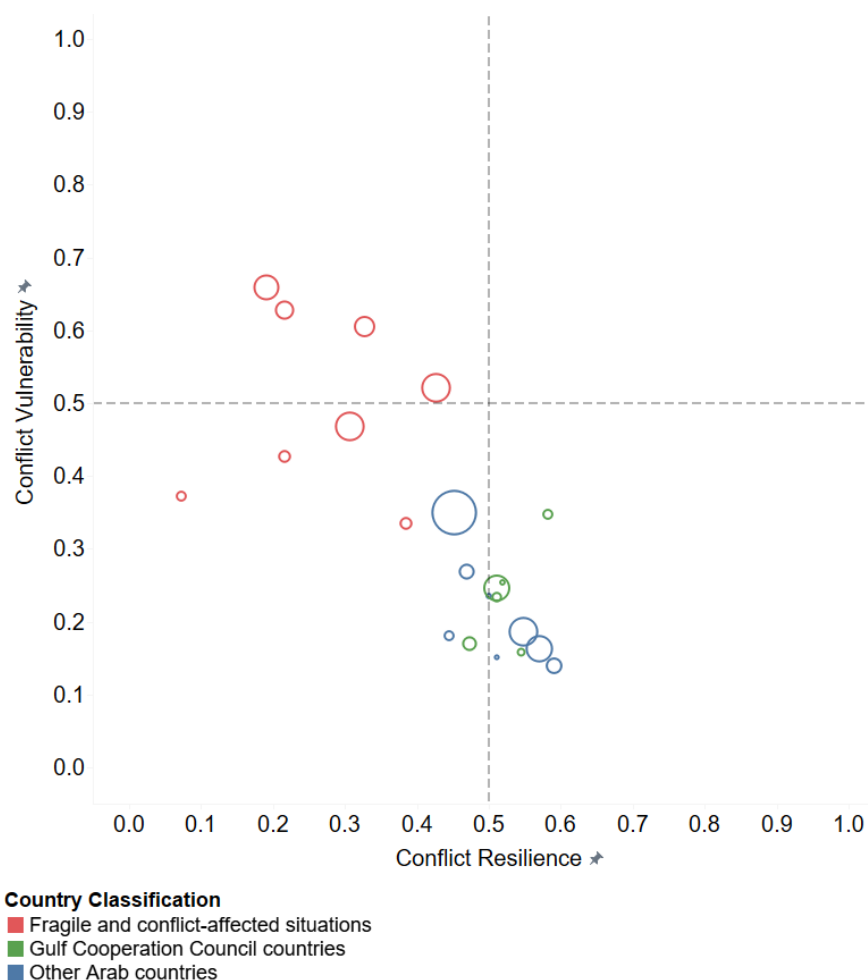
Source: ESCWA elaboration.

Enduring conflict has had a negative impact on the resilience of several countries compared to a decade ago. Notably, nine Arab countries experienced more than a 10 per cent decline in conflict-related resilience since 2010. In contrast, four countries experienced a relative improvement in conflict-related resilience.

Figure 4 illustrates the relationship between conflict vulnerability and resilience among 22 Arab countries in 2021. Conflict vulnerability is represented on the vertical axis, ranging from

"low" to "high" on a scale of 0 to 1. Resilience is plotted on the horizontal axis, also ranging from "low" to "high" on a scale of 0 to 1. The data points are dispersed across the plot, forming four quadrants. By visually depicting the distribution of countries across the scatterplot, we can observe distinct groupings based on their levels of resilience and vulnerability to conflict. The plot highlights the importance of resilience in reducing vulnerability and its potential role in mitigating the risk of conflict among Arab countries.

Figure 4. Conflict vulnerability vs. resilience, 2021



Source: ESCWA elaboration.

The upper left quadrant, which represents the worst scenario, is characterized by countries exhibiting low resilience and high vulnerability to conflict. The countries in this quadrant face significant risks and possess limited capacity to withstand and recover from shocks such as limited territorial control or high numbers of forcibly displaced. Conversely, the lower right quadrant, which represents the best scenario, comprises countries with high resilience and low vulnerability to conflict. These countries demonstrate a stronger capacity to cope and are less prone to conflict-related challenges.

As shown in table 7, a key driver of conflict vulnerability in the Arab region is political instability (0.56 in 2021). Political instability weakens institutions, increases polarization,

deepens societal divisions, erodes trust in the rule of law and can ultimately trigger violence. Of the 22 Arab countries, 5 have a “very high” vulnerability score for the indicator (above 0.80).

Promoting voice and accountability is essential for building resilience and preventing conflict. By providing citizens with access to information and opportunities to participate in decision-making processes, and by holding senior civil servants accountable, institutions can enhance their legitimacy, foster trust and social cohesion, and prevent conflict. In the region, the Voice and Accountability indicator is a key driver of low resilience to conflict (0.29 in 2021) (table 8). Six of the 22 Arab countries have a “very low” resilience score for this indicator (below 0.20).

Table 7. Key risk drivers in the Arab region – conflict vulnerability

Driver	2010	2015	2021
Political Instability	0.52	0.58	0.56
Proliferation of Small Arms	0.47	0.47	0.47
Neighbouring Conflict	0.30	0.63	0.36
Conflict Intensity	0.07	0.19	0.11
Conflict History	0.09	0.20	0.11

Source: ESCWA elaboration.

Table 8. Key risk drivers in the Arab region – conflict resilience

Driver	2010	2015	2021
Military Capacity	0.35	0.31	0.26
Voice and Accountability	0.29	0.30	0.29
Forced Displacement	0.46	0.39	0.36
Territorial Integrity	0.81	0.76	0.79

Source: ESCWA elaboration.

3. The Climate Pathway

A. Climate hazard risk

While climate change does not directly trigger violence, it can influence the likelihood of conflict through a multitude of mechanisms. These include the erosion of livelihoods, increased competition for scarce resources, mass migration and food insecurity. On average, the Climate Hazard Risk domain for the Arab region is ranked as "minor" with a very low level of vulnerability (0.18 in 2021) and a medium level of resilience (0.44 in 2021).

Despite a low average score, the risk posed by climate hazards remains significant in multiple countries, propelled by a surge in extreme weather shocks and an insufficient capacity to cope. Over the past decade, climate hazard vulnerability recorded a deterioration (+59 per cent compared to 2010 levels), caused by the rising human impact of natural disasters. However, the region is increasingly building resilience to climate hazards (+51 per cent compared to 2010 levels) largely due to the adoption of climate adaptation strategies (table 9).

The Arab region is heavily affected by the negative impact of climate change. Droughts and desertification are causing a decline in agricultural production in many areas in the region, while the increasing frequency and intensity of extreme weather and climate events is destroying human settlements and activities, as well as leading to displacements and the emergence of pre-existing issues and grievances.

Climate hazards affect a large share of people in the Arab region with peaks in certain areas due to the rise in the intensity of climate and weather events. Climate hazard vulnerability is assessed by examining the human impact of natural disasters, namely the share of population affected by natural disasters and the number of disaster-induced internal displacements.

Amidst the escalating climate crisis, the Arab region is more vulnerable in 2021 than it was in the previous decade, particularly in three conflict-affected countries, while five other Arab countries have reduced their vulnerability since 2010.

Table 9. Climate hazard risk in the Arab region

Pathway	Risk Domain	Component	2010	2015	2021	Risk Level, 2021	2021 vs. 2010	
Climate	Climate Hazard Risk	Vulnerability	0.11	0.16	0.18	Minor	59%	Deteriorated
		Resilience	0.30	0.35	0.47		51%	Improved

Source: ESCWA elaboration.

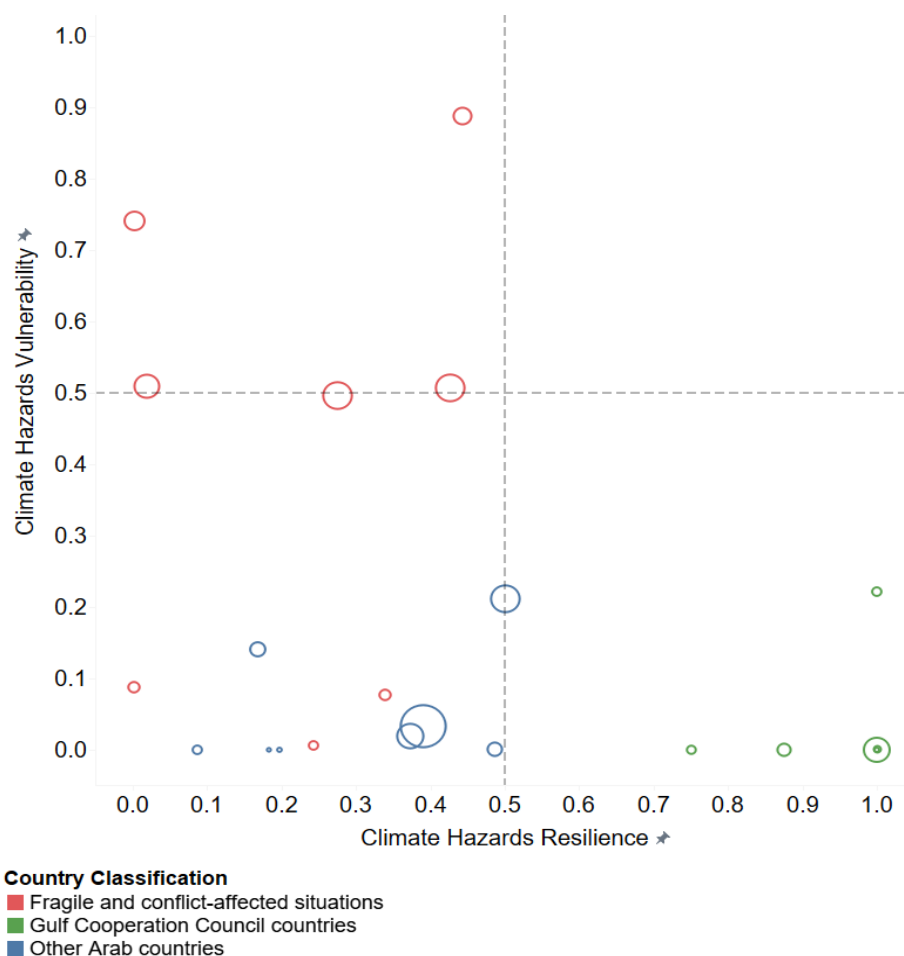
The severity of climate hazards also depends on the capacity to prevent and mitigate the effects of these events, the fragility of socioeconomic and institutional systems, and the direct and indirect effects on conflict in parts of the region. Climate hazard resilience is measured on whether national disaster risk reduction strategies have been adopted and implemented within a country, as well as the availability of climate adaptation-related development finance.

Data limitations exist with respect to this dimension as the dataset for both vulnerability and resilience face scarcity and self-reporting

related issues. Nevertheless, countries with higher economic and institutional capacities are more capable of coping and preventing the negative impact of climate hazards, thus showing lower levels of risk driven by climate hazard vulnerability and higher resilience to cope.

Among the countries for which data is available, three Arab countries experienced a sharp decline in climate hazard resilience over the past decade. In contrast, 8 out of 22 Arab countries experienced an exponential improvement in climate hazard resilience.

Figure 5. Climate hazard vulnerability vs. resilience, 2021



Source: ESCWA elaboration.

Figure 5 showcases the correlation between climate hazard vulnerability and resilience among 22 Arab countries in the year 2021. The vertical axis represents climate hazard vulnerability, which spans from "low" to "high" on a scale of 0 to 1. Resilience is depicted on the horizontal axis, ranging from "low" to "high" on the same scale. The data points are scattered across the plot, forming distinct quadrants which enable country groupings.

The upper left quadrant depicts the most critical scenario, showcasing countries with low resilience and high vulnerability. Within this quadrant, countries face significant climate hazard risks and possess limited adaptation capacities. The lower right quadrant represents the most favourable scenario, comprising countries with high resilience and low vulnerability.

One indicator contributes to the calculation for the climate hazard vulnerability score – the human impact of natural disasters. Extreme

weather events result in extensive damage to key infrastructure and mass displacement, disrupting essential services and livelihoods, leaving populations vulnerable and desperate to meet their basic needs. While the average score for this indicator is "low" (0.18 in 2021) (table 10), it shows a marked deterioration over the last decade (+59 per cent compared to 2010 levels). For the Arab countries for which data is available, one country has a "very high" vulnerability score for this indicator in 2021 (above 0.80).

Financial resources and sound investments are needed to address the climate crises, promote adaptation to the impact of extreme events and enhance the resilience of communities. However, in the Arab region, there is still a limited availability of resources or national capacities dedicated to climate adaptation. The Climate Adaptation Finance indicator is the key driver of low resilience in the region (0.33 in 2021) (table 11). "Very low" resilience levels (below 0.20) are evident in 14 of the 22 Arab countries.

Table 10. Key risk drivers in the Arab region – climate hazard vulnerability

Driver	2010	2015	2021
Disaster Impact	0.11	0.16	0.18

Source: ESCWA elaboration.

Table 11. Key risk drivers in the Arab region – climate hazard resilience

Driver	2010	2015	2021
Adaptation Finance	0.31	0.35	0.33
Adaptation Strategies	0.29	0.36	0.70

Source: ESCWA elaboration.

B. Natural resource risk

Several Arab states rely heavily on agriculture – which is highly vulnerable to climate change – as a source of income. The lack of water is a pressing issue and a threat to the region’s sustainability. The control of natural resources can ignite conflicts between different actors. In areas where water is scarce, poor management practices, lack of conservation and inefficiencies can also create social tensions. The Natural Resource Risk domain is ranked as “significant” with a “medium” level of vulnerability (0.49 in 2021) and a “low” level of resilience (0.38 in 2021) (table 12).

The link between scarcity of renewable resources such as water and land, and conflict is complex. Pre-existing environmental vulnerabilities and scarcities can increase exposure to climate impact or reduce the ability to cope and affect the transmission of risk through the aforementioned transmission mechanisms. Natural resource vulnerability is “very high” or “high” in four countries. Low reliance on agriculture and water scarcity can indicate lower vulnerability to adverse climate impact in the Arab region.

There are only three countries that are less vulnerable, relatively, for the natural resource dimension. Fourteen countries are more

vulnerable in 2021 than they were in the previous decade. On the other hand, only 3 countries out of 22 Arab countries show reduced vulnerability compared to 2010 levels.

As the demand for water increases, particularly in rapidly urbanizing areas and for agricultural purposes, the withdrawal of freshwater from rivers, lakes and groundwater sources can exceed the natural replenishment rate, leading to the depletion of these resources. Poor water management practices, lack of water conservation, and inefficient use and allocation can cause socioeconomic challenges in areas where water is scarce. Natural resource resilience is “very low” or “low” in 14 out of 22 countries. Well-functioning, sustainable and fair resource management, governance and dispute settlement processes can build resilience and reduce risk. Natural resource resilience is recorded as “high” in one country.

The scarcity of natural resources had a negative impact on the resilience of several countries compared to a decade ago. Notably, three countries have experienced a more than 10 per cent decline in natural resource resilience since 2010. In contrast, some countries experienced a relative improvement in natural resource resilience, three at more than 5 per cent compared to 2010.

Table 12. Natural resource risk in the Arab region

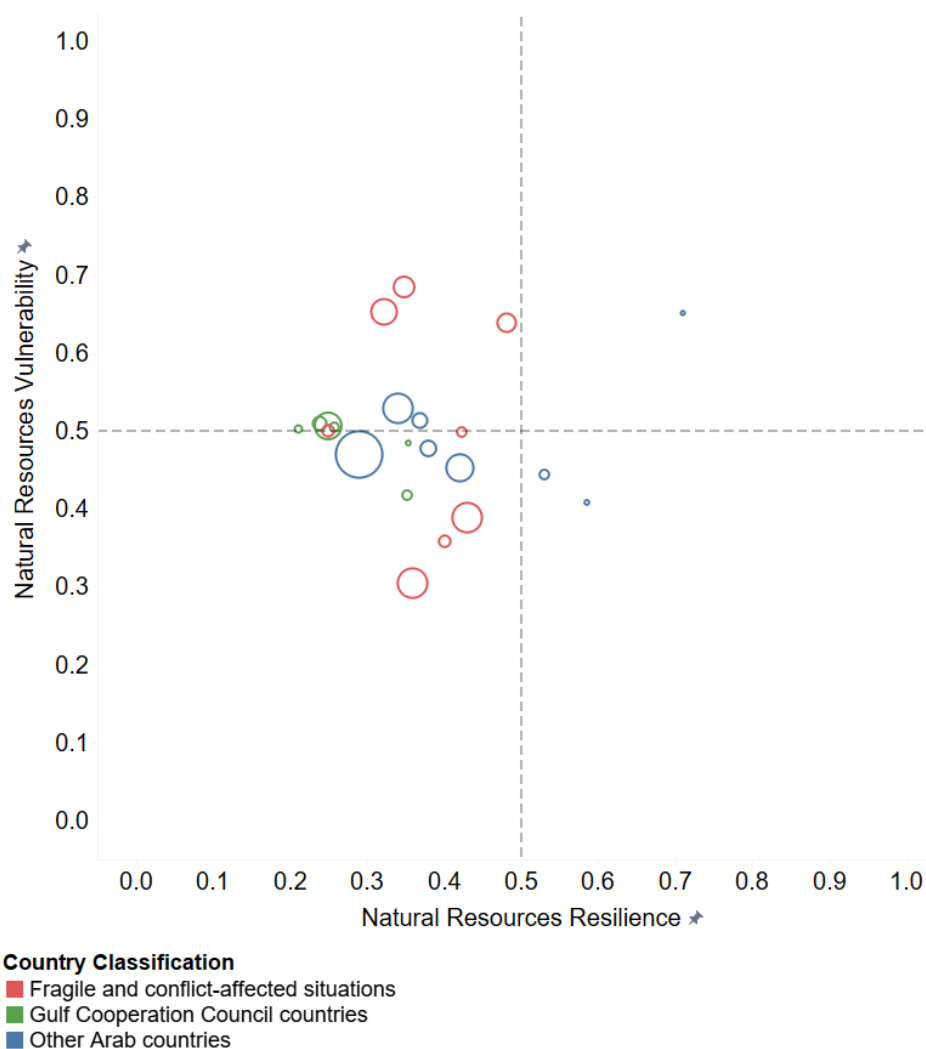
Pathway	Risk Domain	Component	2010	2015	2021	Risk Level, 2021	2021 vs. 2010	
Climate	Natural Resource Risk	Vulnerability	0.47	0.49	0.49	Significant	5%	Deteriorated
		Resilience	0.39	0.39	0.38		-4%	Deteriorated

Source: ESCWA elaboration.

Figure 6 illustrates the relationship between natural resource vulnerability and resilience among 22 Arab countries in 2021. Natural resource vulnerability is represented on the vertical axis, ranging from "low" to "high" on a scale of 0 to 1. Resilience is plotted on the horizontal axis, also ranging from "low" to

"high" on a scale of 0 to 1. The data points are dispersed across the plot, forming four quadrants. By visually depicting the distribution of countries across the scatterplot, we can observe distinct groupings based on their levels of resilience and vulnerability.

Figure 6. Natural resource vulnerability vs. resilience, 2021



Source: ESCWA elaboration.

The upper left quadrant, which represents the most critical scenario, is characterized by countries exhibiting low resilience and high vulnerability. Within this quadrant, countries – mainly conflict-affected countries – face substantial risks related to natural resources and possess limited abilities to endure and rebound from shocks, including land or water stress. The lower right quadrant, representing the most favourable scenario, consists of countries with high resilience and low vulnerability.

Water availability is an ever-increasing challenge in the Arab region, posing an existential threat. Climate change is exacerbating the water crisis with irregular rainfall patterns leading to more intense droughts, floods, and other extreme climate and weather events. Control over freshwater

resources can fuel tensions and conflict between groups and communities. Water Scarcity is a key driver of vulnerability in the Arab region (0.79 in 2021) (table 13). Out of 22 Arab countries, 9 display a “very high” vulnerability score (above 0.80) for the indicator.

As the demand for water increases, particularly in rapidly urbanizing areas and for agricultural purposes, the withdrawal of freshwater from rivers, lakes and groundwater sources can exceed the natural replenishment rate, leading to its depletion. Poor water management practices, lack of water conservation, and inefficient use and allocation can trigger social tensions in areas where water is scarce. The Water Stress indicator, a key driver of low resilience (0.26 in 2021) (table 14), reveals that in 8 out of the 22 Arab countries, the level is “very low” (below 0.20).

Table 13. Key risk drivers in the Arab region – natural resource vulnerability

Driver	2010	2015	2021
Water Scarcity	0.75	0.78	0.79
Reliance on Agriculture	0.17	0.19	0.18

Source: ESCWA elaboration.

Table 14. Key risk drivers in the Arab region – natural resource resilience

Driver	2010	2015	2021
Water Stress	0.26	0.26	0.26
Land Stress	0.52	0.52	0.50

Source: ESCWA elaboration.

4. The Development Pathway

A. Economic risk

Conflict increases a country's economic vulnerability, perpetuating the risk of prolonged conflict as well as worsening the economic capacity of a country to recover from the "conflict trap". Conversely, a strong economy helps alleviate poverty and reduce tensions. On average, the Economic Risk domain for the Arab region is ranked as "moderate" with a medium level of vulnerability (0.43 in 2021) and a medium level of resilience (0.55 in 2021).

Over the last decade, the Arab region saw a deterioration in the Economic Risk domain (+9 per cent vulnerability and -15 per cent resilience compared to 2010 levels). Amidst increased food insecurity, reliance on foreign trade and capital markets, and a narrowing fiscal space, Arab nations are more vulnerable to the impact of shocks than they were in the previous decade.

The economy of the Arab region is characterized by disparities. Oil and gas exports are the

primary source of revenue for some countries, accounting for a substantial portion of their gross domestic product (GDP). Conflict-affected countries face sluggish growth, mounting debt, high unemployment rates and rising income inequality. Conflict and instability worsen the economic outlook in certain countries, leading to a significant contraction in economic activities and exacerbating existing challenges. Economic-related vulnerability is "low" in 9 out of 22 Arab countries.

Countries with a poor capacity to mobilize domestic resources, particularly conflict-affected countries, are more vulnerable in 2021 than they were in the previous decade. Economic-related vulnerability has also increased in countries not currently experiencing a violent conflict but who are bordering an active conflict. Six countries have reduced their vulnerability since 2010. The increasing trend in economic vulnerability is mainly due to rising food insecurity, financial dependency and the rise in inequality in fragile countries in the region.

Table 15. Economic risk in the Arab region

Pathway	Risk Domain	Component	2010	2015	2021	Risk Level, 2021	2021 vs. 2010	
Development	Economic Risk	Vulnerability	0.39	0.40	0.43	Moderate	9%	Deteriorated
		Resilience	0.65	0.59	0.55		-15%	Deteriorated

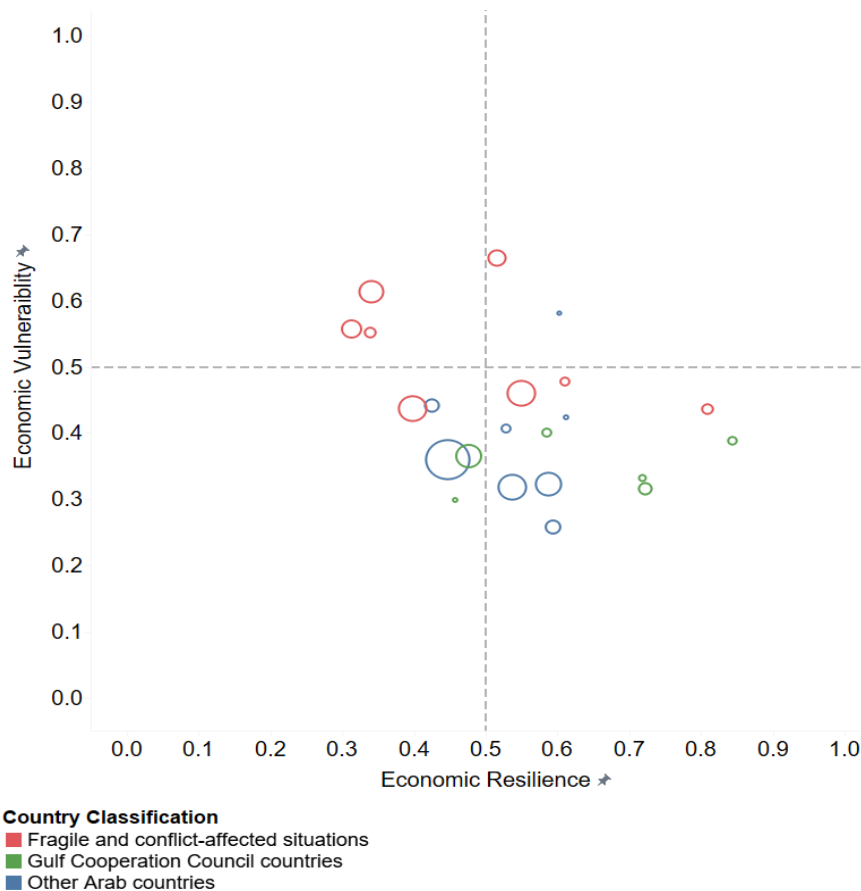
Source: ESCWA elaboration.

The risk of conflict arising from economic factors is not only influenced by an economy's vulnerability but also the degree at which it can withstand external shocks. Having adequate fiscal space enables governments to maintain macroeconomic stability. Countries that are able to access capital markets can mitigate the negative impact of crises, reducing the likelihood of social tensions and the potential for conflict. Without proper debt management policies, however, they might be vulnerable to external shocks and thereby increasing social tensions and falling into conflict. Economic-related resilience was "very high" or "high" in seven Arab countries in 2021. A weak economy hinders a government's ability to respond

effectively to economic challenges such as recessions, high unemployment rates or inflation. This can lead to increased economic instability, inequality and social unrest, potentially escalating into conflict. Economic-related resilience is "low" in three countries.

Economic downturns have had a negative impact on the resilience of several countries compared to a decade ago. Notably, 13 countries have experienced more than a 10 per cent decline in economic-related resilience since 2010. Conversely, four countries experienced a relative improvement in economic-related resilience compared to 2010.

Figure 7. Economic vulnerability vs. resilience, 2021



Source: ESCWA elaboration.

Figure 7 exhibits the relationship between economic vulnerability and resilience among 22 Arab countries in 2021. The vertical axis represents the level of economic vulnerability, ranging from "low" to "high" on a scale of 0 to 1. The horizontal axis portrays the level of resilience, spanning from "low" to "high" on the same scale. The data points are dispersed across the plot, resulting in distinct quadrants that facilitate country groupings.

The upper left quadrant depicts the worst-case scenario where countries exhibit low resilience and high vulnerability. Countries within this quadrant face significant vulnerabilities such as reliance on food imports, and possess limited resilience such as a lack of fiscal space. The lower right quadrant represents the best-case scenario, comprising countries with a stronger and more resilient economy.

As shown in table 16, the main driver of high economic vulnerability in the Arab region is the

Food Insecurity indicator (0.65 in 2021). When people do not have access to sufficient, nutritious food it can lead to growing grievances within the community. In addition, over-reliance on food resources traded on international markets can increase vulnerability to price shocks caused by environmental factors. Demonstrations, riots and conflict are more likely when international and domestic food prices rise. Among the 22 Arab countries, 7 display a "very high" vulnerability score (above 0.80) for the Food Insecurity indicator.

For national governments to invest in socioeconomic development, governance and conflict prevention, adequate financial resources are needed. In the Arab region, the domestic availability of funds is often scarce and constrained by significant levels of debt. Inadequate Fiscal Space is the main driver of low resilience in the Arab region (0.39 in 2021) (table 17). "Very low" resilience levels (below 0.20) are prevalent in 2 of the 22 Arab countries.

Table 16. Key risk drivers in the Arab region – economic vulnerability

Driver	2010	2015	2021
Food Insecurity	0.56	0.59	0.65
Inequality	0.46	0.43	0.42
Trade Dependence	0.40	0.40	0.40
Financial Dependence	0.15	0.18	0.24

Source: ESCWA elaboration.

Table 17. Key risk drivers in the Arab region – economic resilience

Driver	2010	2015	2021
Fiscal Space	0.62	0.53	0.39
Economic Development	0.61	0.60	0.59
Economic Growth	0.72	0.64	0.66

Source: ESCWA elaboration.

B. Social risk

Inadequate social systems can deepen inequalities and social divisions, creating a fertile ground for tensions. Poor health-care systems and insufficient social protection leave vulnerable populations without necessary support, particularly during times of crises. Strong social support structures and comprehensive social policies contribute to peace and stability.

As shown in table 18, the Social Risk domain is ranked as "minor" with a low level of vulnerability (0.33 in 2021) and a medium level of resilience (0.47 in 2021). It is important to note that the regional average can conceal the context-specificity of social risks, which can vary substantially across countries. While challenges and disparities persist, regional Social Risk has witnessed an improvement over the past decade (-9 per cent vulnerability and +22 per cent resilience compared to 2010 levels) with expanding social safety nets and decreasing infant mortality rates.

A large population of young people without adequate employment and educational opportunities can breed disillusionment and social marginalization. Social vulnerability is "high" or "medium" in 10 countries. Stronger social cohesion is exhibited in countries with

better living conditions and low infant mortality rates. In the Arab region, six countries are at a "very low" level of vulnerability for society dimensions.

Due to the persistent challenges, conflict-affected countries are more vulnerable in 2021 than they were in the previous decade. Social vulnerability also increased in countries not currently experiencing violent conflict, although 11 countries have reduced their vulnerability by more than 10 per cent since 2010.

The risk of conflict stemming from social factors is not solely determined by vulnerability but also by the degree at which a society can withstand external shocks. Accessible and comprehensive health coverage, strong social protection systems, quality education and access to clean water and adequate sanitation facilities are crucial for social well-being. When communities have access to quality social services, it fosters stability and cooperation. Social resilience was "very high" or "high" in 10 countries in 2021 and "very low" in 3 countries.

Income inequality and inequality in accessing basic services had a negative impact on the resilience of several countries compared to a decade ago. In contrast, most of the Arab countries experienced a relative improvement in society resilience.

Table 18. Social risk in the Arab region

Pathway	Risk Domain	Component	2010	2015	2021	Risk Level, 2021	2021 vs. 2010	
Development	Social Risk	Vulnerability	0.37	0.34	0.33	Minor	-9%	Improved
		Resilience	0.50	0.53	0.51		4%	Improved

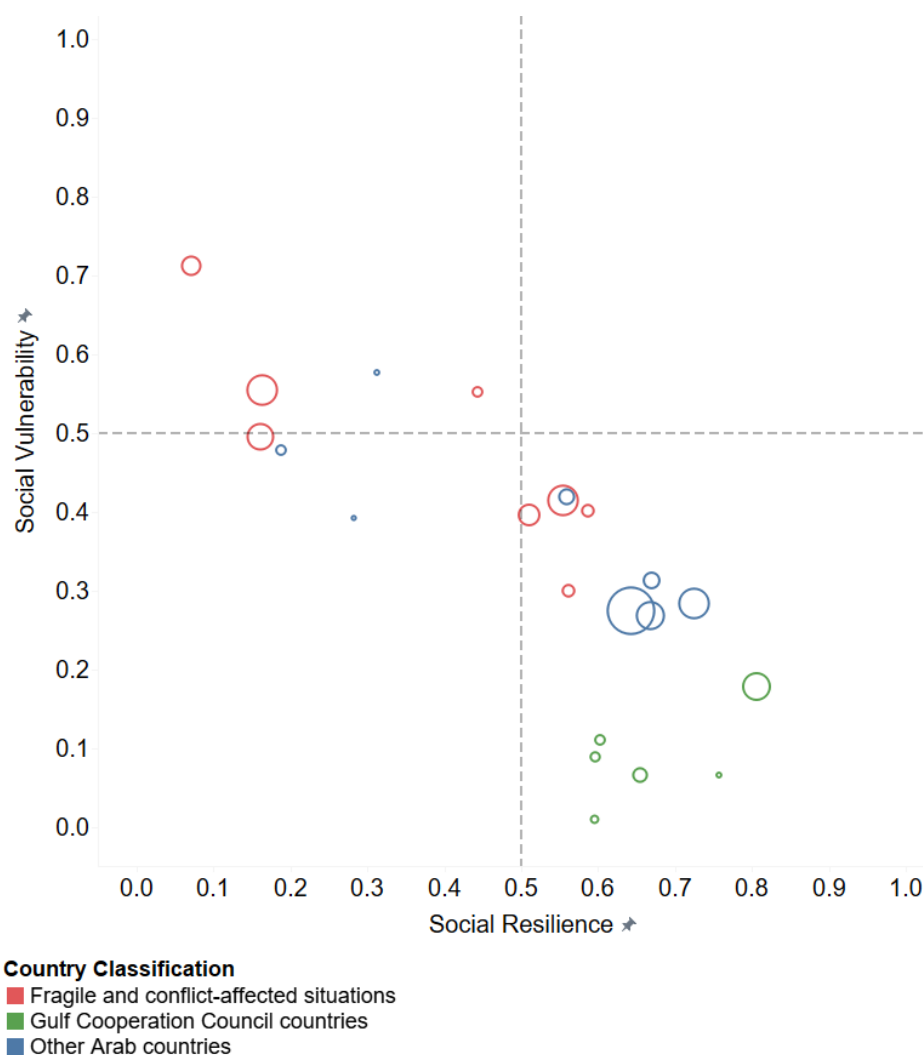
Source: ESCWA elaboration.

Figure 8 shows the relationship between social vulnerability and resilience in the Arab region for 2021. The vertical axis represents the level of social vulnerability ranging from "low" to "high" on a scale of 0 to 1. The horizontal axis portrays the level of resilience spanning from "low" to "high" on the same scale.

The upper left quadrant depicts the worst-case scenario, where countries exhibit low resilience

and high vulnerability. Countries within this quadrant face challenges, including unemployment rates and limited social protection coverage. The lower right quadrant represents the best-case scenario, encompassing countries with higher social resilience and low vulnerability. These countries demonstrate favourable indicators such as improved education levels, better health coverage and low infant mortality rates.

Figure 8. Social vulnerability vs. resilience, 2021



Source: ESCWA elaboration.

Table 19. Key risk drivers in the Arab region – social vulnerability

Driver	2010	2015	2021
Unemployment	0.40	0.44	0.49
Youth Bulge	0.51	0.42	0.36
Infant Mortality	0.19	0.17	0.15

Source: ESCWA elaboration.

Table 20. Key risk drivers in the Arab region – social resilience

Driver	2010	2015	2021
Social Protection		0.36	0.26
Health Coverage	0.32	0.36	0.41
Education	0.48	0.51	0.55
Water and Sanitation Services	0.69	0.74	0.77

Source: ESCWA elaboration.

The main driver of high vulnerability is the Unemployment indicator (0.49 in 2021) (table 19). Within the 22 Arab countries, 5 are classified as "very high" (exceeding 0.80) in 2021.

Social protection is essential for building resilience and preventing conflict. Social safety nets provide vulnerable communities with the resources needed to cope with shocks, reducing tensions and promoting social cohesion. Insufficient social protection is the primary cause of weak resilience in the Arab region (at an alarming 0.26 in 2021) (table 20). Among the Arab countries with available data, nine have a "very low" resilience (below 0.2) score for the Social Protection indicator.

C. Institutional risk

Ineffective institutions can create a breeding ground for social tensions and conflict. Corruption can erode trust in institutions and

undermine the legitimacy of the state. Without effective mechanisms for participating in decision-making processes and holding civil servants accountable, people may resort to extreme measures, including violent conflict, to demand change. The Institutional Risk domain is ranked as "significant" with a high level of vulnerability (0.70 in 2021) and a medium level of resilience (0.40 in 2021) (table 21). Over the last decade, the Arab region saw a deterioration in the Institutional Risk domain (-9 per cent vulnerability and +22 per cent resilience compared to 2010 levels).

Weak institutions marked by corruption and governance failures contribute to an environment of injustice. As a result, societal groups may develop grievances as they perceive institutions as serving the few and not all citizens as they are mandated. Institutional vulnerability is "very high" in 8 out of 22 Arab countries. Controlling corruption helps mitigate the risk of conflict by promoting transparency, trust and social cohesion.

Table 21. Institutional risk in the Arab region

Pathway	Risk Domain	Component	2010	2015	2021	Risk Level, 2021	2021 vs. 2010	
Development	Institutional Risk	Vulnerability	0.66	0.69	0.70	Significant	7%	Deteriorated
		Resilience	0.43	0.41	0.40		-8%	Deteriorated

Source: ESCWA elaboration.

The risk of conflict arising from failing institutions is not only shaped by the control of corruption. Effective institutions and a strong rule of law mitigate the risk of conflict by providing a framework for fair dispute resolution, promoting social order, and holding individuals and institutions accountable for their actions. Institutional resilience is “high” in four Arab countries. When institutions fail to provide essential services and meet the needs of the population such as access to justice, basic infrastructure and public goods, it can breed frustration and dissatisfaction. Institutional resilience is classified as “very low” in four countries and “low” in eight countries.

Weak rule of law and government ineffectiveness have had a negative impact on the resilience of several countries compared to a decade ago. Notably, seven countries experienced more than a 10 per cent decline in institutional resilience since 2010. In contrast, four countries experienced a more than 10 per cent improvement in institutional resilience since 2010.

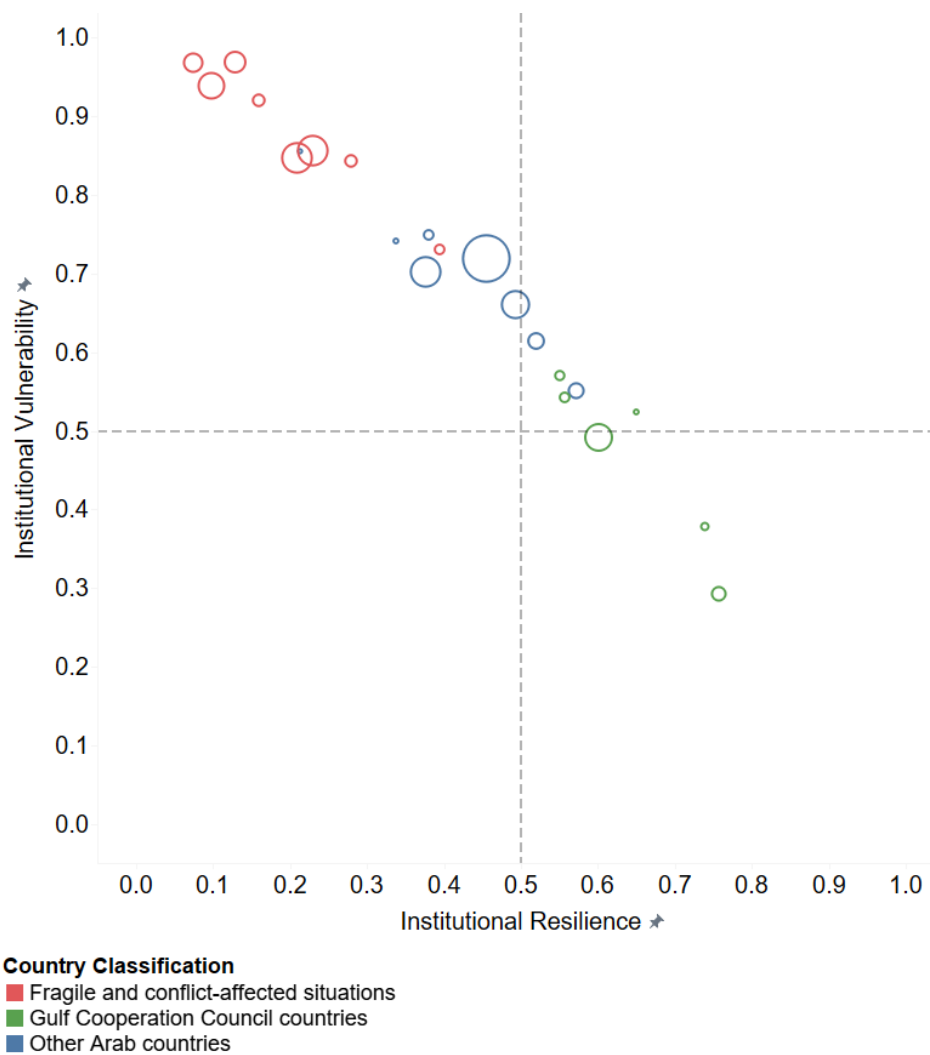
Figure 9 illustrates the correlation between institutional vulnerability and resilience in the Arab region for the year 2021. The vertical axis represents the degree of institutional vulnerability ranging from “low” to “high”

on a scale of 0 to 1. The horizontal axis portrays the level of resilience ranging from “low” to “high” using the same scale. The data points result in distinct quadrants that facilitate country groupings.

The upper left quadrant depicts the worst-case scenario, where countries exhibit low resilience and high vulnerability. Countries within this quadrant face challenges, including lack of control of corruption and limited government effectiveness. The lower right quadrant represents the best-case scenario, encompassing countries with higher institutional resilience and low vulnerability. These countries demonstrate favourable indicators such as higher control on corruption, more efficient governments and implementation of rule of law.

In the Institutional risk domain, corruption is a significant threat to peace and stability and a key driver of high vulnerability in the Arab region (0.70 in 2021) (table 22). As public resources are diverted, this can create social and political tensions, especially in the presence of marginalized groups. Corruption can erode trust in institutions and undermine the legitimacy of the state. Out of 22 Arab countries, 8 display a “very high” vulnerability score (above 0.80) for the Corruption indicator.

Figure 9. Institutional vulnerability vs. resilience, 2021



Source: ESCWA elaboration.

Table 22. Key risk drivers in the Arab region – institutional vulnerability

Driver	2010	2015	2021
Corruption	0.66	0.69	0.70

Source: ESCWA elaboration.

Inadequate government effectiveness stands as the primary factor behind the low level of resilience in the Arab region, scoring 0.38 in

2021 (table 23). Six Arab countries experienced "very low" levels of resilience (below 0.20).

Table 23. Key risk drivers in the Arab region – institutional resilience

Driver	2010	2015	2021
Government Effectiveness	0.41	0.39	0.38
Rule of Law	0.45	0.43	0.42

Source: ESCWA elaboration.

5. Evolution of the risk landscape in the Arab region: a comparative assessment

A. Ranking of risk domains in the Arab region

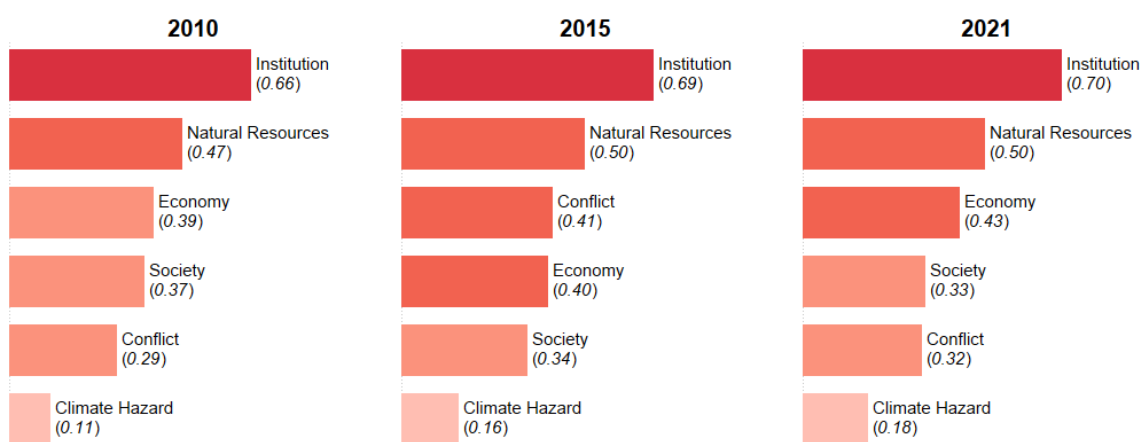
The figures included in this section illustrate a comparative ranking of the risk domains in the Arab region for the years 2010, 2015 and 2021. The ranking offers insights on the relative importance and position of the six risk domains – conflict, natural resource, climate hazard, economic, social and institutional – for both vulnerability and resilience. By examining the

data, it is possible to gain insights into the changing dynamics and significance of risk for the region over the past decades.

Figure 10 presents the ranking of vulnerability domains based on their relative position to each other.⁴ Institutional vulnerability emerges as the primary driver of risk in all three considered years. Among the drivers of high levels of institutional vulnerability are weak or inexistent accountability and control of corruption.

Figure 10. List of domains ranked by highest to lowest vulnerability

Vulnerability



Source: ESCWA elaboration.

⁴ The figure shows the placement of the domains in relation to each other. This means that higher rankings of certain domains do not necessarily indicate high or low vulnerability. The difference between rankings may be minimal. For example, while natural resource vulnerability is ranked as the second highest risk driver, it is still classified at a medium level of vulnerability. However, it is important to acknowledge that the risk induced by natural resource vulnerability is relatively higher compared to some of the other domains.

Natural resource vulnerability is another important driver of risk reflecting the structural issues surrounding water availability, its management – or rather mismanagement – and the reliance on agriculture in several Arab states. Economic vulnerability is also among the three main drivers of risk in both 2010 and 2021 for reasons that include conflict and its spillover effects, in addition to the impact of the COVID-19 induced crises, mainly but not exclusively, increasing food insecurity in conflict-affected countries.

Conflict vulnerability was among the top risk domains in 2015, corresponding with the peak of violence recorded the previous year but not in 2021. This dimension is especially relevant for conflict-affected countries including those that have had to cope with an influx of refugees and other conflict spillover effects. Conflicts have also diverted resources from development to the security sector while creating an unfriendly foreign investment climate.

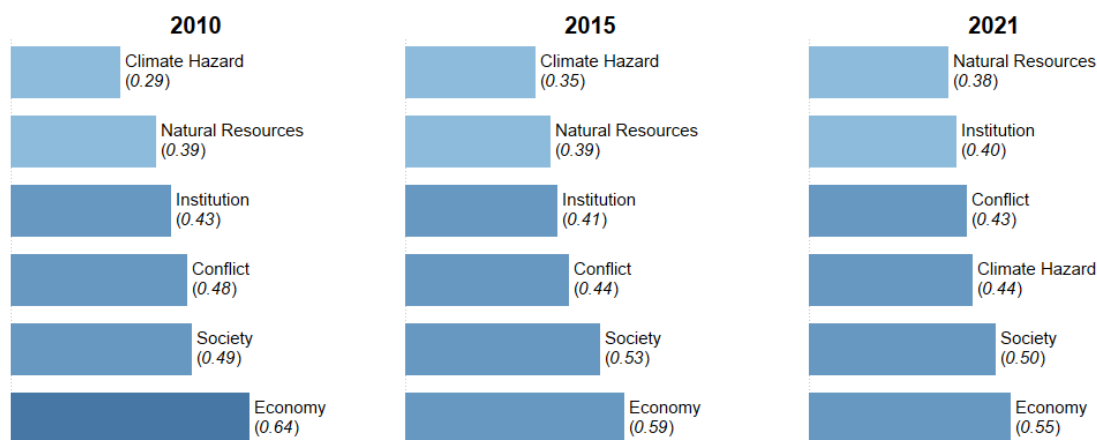
With regard to resilience, figure 11 shows that economy is ranked relatively high in the three years considered, while climate-related

resilience ranks among the lowest. While economic resilience has been decreasing throughout the years, it still remains among the highest in terms of resilience when compared to the other domains. This could be attributed mainly to the cushioning effects of the resource-rich Arab states, and the ripple effects that fossil fuel revenues bring to the Arab region.

In 2010, social resilience ranked second lowest in the Arab region. This may be explained by the mass social movements that took place in several Arab countries in 2011 which evolved into violent conflicts in some states. Although conflict resilience was ranked as the second highest resilience by domain in 2010, by 2021 it decreased to be among the lowest. Weak resilience to conflict is mainly attributed to the legacies of conflict including lack of state authority over the territory, in addition to the high prevalence of refugees and internally displaced persons (IDPs) that continues to linger on to the present day. Refugees and IDPs' inability to return to their home is indicative of unresolved political, security and socioeconomic challenges caused by conflict.

Figure 11. List of domains ranked by lowest to highest resilience

Resilience



Source: ESCWA elaboration.

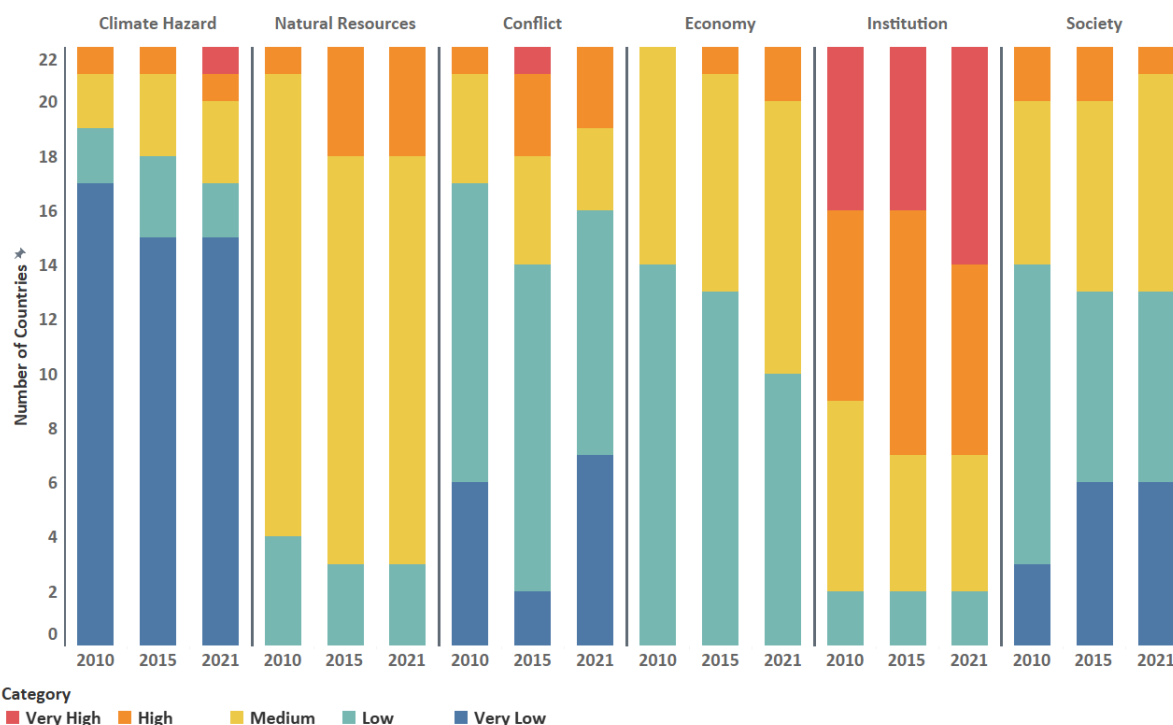
B. Distribution of Arab countries across risk domains

The figures presented in this section illustrate the distribution of Arab countries across the five levels of vulnerability and resilience (“very high”, “high”, “medium”, “low” and “very low”). The data provides a visual representation of the number of countries falling into each vulnerability and resilience level, offering insights into the overall risk landscape in the region.

Figure 12 shows the number of countries in each domain with the associated level of vulnerability ranging from “very low” to “very high”. The six domains are presented for the three points in time considered – 2010, 2015 and 2021.

As shown in figure 12, institutional vulnerability has been the main driver of risk since 2010 with eight countries classified as being at a “very high” level of institutional vulnerability and six countries at “high” institutional vulnerability. Over the span of a decade, the number of countries with a “high” and “very high” level of institutional vulnerability increased, signalling a deterioration in the condition of institutions and governance in the region. This can be explained by wars and political tensions in several countries, as well as a significant weakness in the implementation of rule of law. Institutional vulnerability remains one of the primary causes – if not the main cause – of risk exposure in the Arab region.

Figure 12. Vulnerability level of countries by domain and year

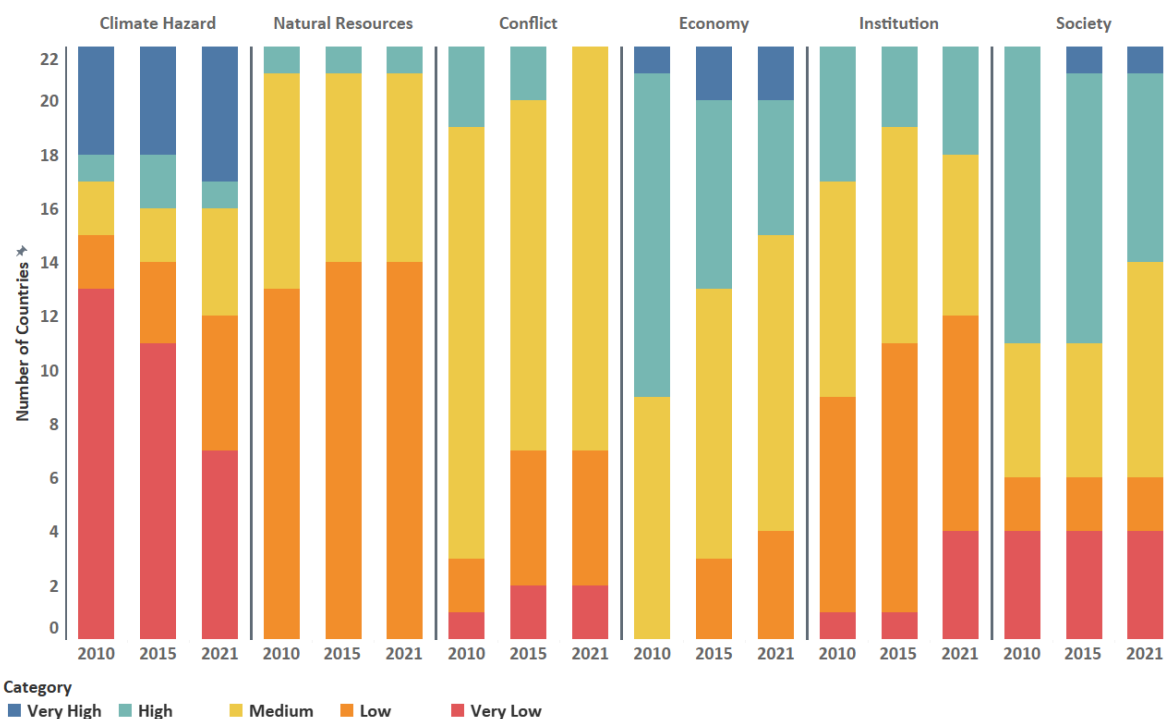


Source: ESCWA elaboration.

The number of countries classified as “high” or “medium” in economic vulnerability has increased from 8 countries in 2010 to 11 countries in 2021. In terms of social vulnerability, eight countries range between “high” and “medium”. For both climate hazard vulnerability and natural resource vulnerability, it can be observed that an increasing number of countries are becoming more vulnerable to climate-related impact. By 2021, five countries were classified as facing “very high”, “high” or “medium” climate hazard vulnerability, while 18 countries were at “high” or “medium” natural resource vulnerability.

Between 2010 and 2021, conflict vulnerability peaked in 2015 with eight countries in the “very high”, “high” or “medium” categories. In 2021, conflict vulnerability dropped to six countries. This could be explained by a reduction in battle related deaths that could be attributed to battle fatigue among combatants, a changing geopolitical dynamic or ongoing mediation efforts. Still, the risk that political processes or the several “no peace, no war” status quos existing in several Arab countries could rapidly deteriorate into violence remains a reality. Events in the Sudan since April 2023 demonstrate that even countries at “medium risk” can surreptitiously plunge into conflict.

Figure 13. Resilience level of countries by domain and year



Source: ESCWA elaboration.

Figure 13 shows the number of countries in each domain with the associated level of resilience ranging from “very low” to “very high”. Climate hazard resilience remains the lowest with 12 countries categorized as either “very low” or “low” in 2021. The fact that more than half of Arab States are within the “low” and “very low” resilience category is worrisome and could be attributed to weak institutional as well as financial capacities, in addition to the impact of conflict on people, society and institutions. Fifteen countries have a “low” level of natural resource resilience as the challenges of land and water stresses remain unaddressed. “Low” resilience in the Climate Pathway indicates that the implementation of policies and institutional capacities to cope with climate-related impact and the deterioration of already scarce natural resources requires further diligence.

One of the main drivers of risk for the period considered is the lack of institutional resilience. Figure 14 shows that institutional resilience has decreased over time with more than half the countries recording “low” or “very low” levels of resilience. Despite the slight reduction in the number of countries at “low” or “very low” levels of social resilience, it remained a significant driver of risk in the Arab region in 2021. Economic resilience decreased over time as economic conditions deteriorated in several countries due to domestic conflict or regional instability, as well as the impact of economic trends and shocks such as the crisis induced by the COVID-19 pandemic.

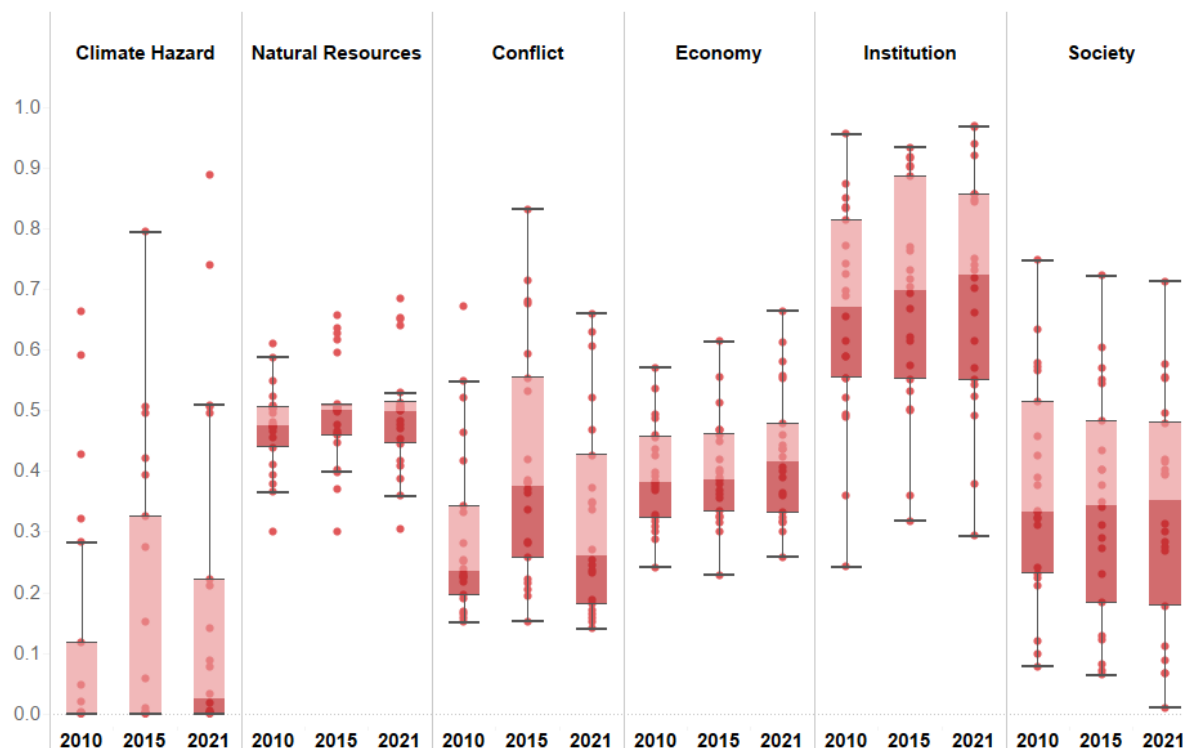
Most of the countries in 2021 had a “medium”, “low” or “very low” level of conflict resilience driven by weak rule of law and government

ineffectiveness. Since 2010, countries at “very low” or “low” resilience to conflict have increased from four to seven countries. It is of note that since 2015, when intensity of conflict was at its highest, the number of countries at the “low” or “very low” resilience category has not changed to the present day.

The results presented above offer an indication as to which areas of risk are of particular concern in the Arab region. Institutional risk emerged as a key risk factor. Weak institutions that are marred by corruption, weak rule of law and government ineffectiveness will undermine people and societies’ ability to address vulnerabilities that are associated with conflict, climate change and de-development. Their resilience to withstand the pressures stemming from past conflicts, political instability or neighbouring conflicts, as well as the proliferation of arms, will diminish dramatically. Institutional risk will also weaken the resilience of people and society to face the vulnerabilities related to trade and financial dependency, food insecurity and socioeconomic inequality.

Figure 14 presents the distribution of data and the median for Arab countries at the domain level for the vulnerability dimension. It shows how the data is distributed and highlights any outliers, as well as the variation in the median and the range at which the data is clustered. The climate vulnerability dimension presents two different types of distribution: natural resource vulnerability, which is the most concentrated in terms of distribution of countries around the regional median; and climate hazard vulnerability, the distribution of which shows the largest spread of volatility to climate and weather-related events in the region.

Figure 14. Distribution and median by domain and by year for the vulnerability dimension



Source: ESCWA elaboration.

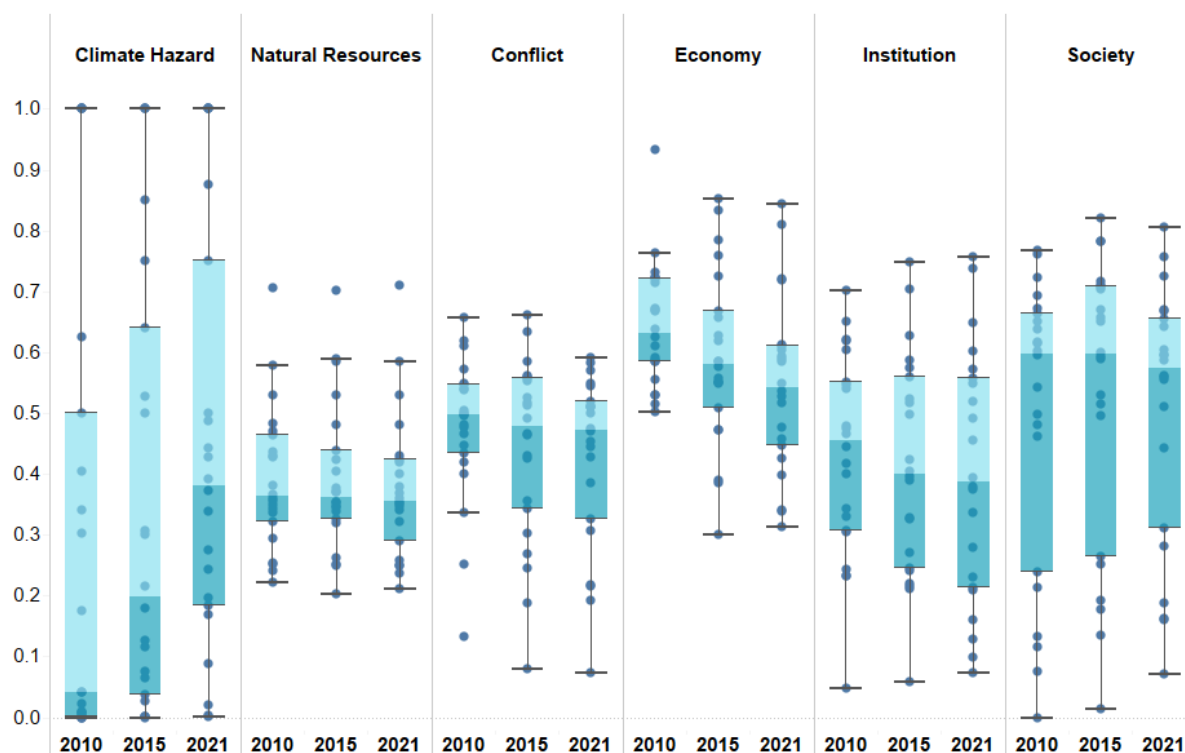
Another dimension whose distribution increased, especially in 2015 with the peak of violence in the Arab region, is conflict. Countries in the region experienced relatively “high” vulnerability to conflict in 2015, while others were only partially affected. Looking at the distribution of countries in the Development Pathway, large distributions can be observed for the social and institutional vulnerability dimensions, with minor changes throughout the years. However, the economic vulnerability dimension shows that the distribution of data is not significantly wide and that many countries are concentrated around the median.

Looking at the trend of the medians, it can be observed that vulnerability in the region has increased in all six dimensions since 2010. The

only outlier is conflict whereby vulnerability surged in 2015 then dropped to levels that were still higher than in 2010. Nevertheless, this increasing trend of vulnerability will continue its upward trajectory and will ultimately increase the overall regional risk of conflict if no preventative measures are undertaken by local, national, regional and international actors.

Figure 15 presents the distribution of data and the median for Arab countries at the domain level for the resilience dimension. Looking at the distribution of countries by domain, climate hazard resilience has the widest distribution while its median has significantly increased since 2010. An increasing dispersion of countries is reported over time for almost all dimensions but mainly for conflict, institutions and society.

Figure 15. Distribution and median by domain and by year for the resilience dimension



Source: ESCWA elaboration.

Only climate hazard and social resilience show an increasing trend over the period considered, while all other dimensions show decreasing trends. The median of the climate hazard resilience dimension has significantly increased since 2010, while the rise in social resilience was limited. For climate hazard resilience, this is mainly due to an increase in adaptive strategy implementation and to the slight increase in adaptation finance. For social resilience, the increase is due to the progressive increase in all four drivers, despite the impact of conflict in affected countries.

The regional median has decreased for institutional, economic, natural resource and

conflict resilience since 2010. The median for economic and institutional resilience is particularly alarming as they both experienced a sharp reduction over the considered period. Economic resilience decreased from “high” to “medium” levels, while institutional resilience decreased from “medium” to “low” levels. All the drivers for institutional and economic resilience decreased over time, particularly public debt which became a prominent concern in many countries. The medium to long-term effects of conflict in countries affected directly or indirectly by it, as well as other mega trends at country, regional and international levels such as changes in the value chains of the production and distribution of goods and

services and the socioeconomic shocks caused by external factors and agents, are all potential drivers of risk for the Arab countries.

Natural resource, climate hazard and institutions are ranked among the lowest resilience dimensions when compared to other domains. This is particularly concerning as institutions

and natural resources also present “medium” and “high” levels of vulnerability, respectively. If the decrease in resilience remains in coming years, the overall regional risk will increase accordingly, especially if no preventive measures are undertaken by local, national, regional and international stakeholders to strengthen resilience.

Annex. Data corrigendum

This annex aims at providing all the changes that were implemented in the list of drivers included in the methodological paper *Arab Risk Monitor: Quantifying the drivers of risk of conflict, version 1.0*.⁵ Changes have been applied to include additional drivers to the analysis of risk, as well as to better reflect reality in the Arab region given data availability constraints especially for certain groups of countries. The changes are presented following the structure of the table.

List of drivers and variables

Pathway	Domain	Dimension	Driver	Variable
Conflict Pathway	Historical Grievances	Vulnerability	Conflict Intensity	Battle-related deaths per 100,000 people (SDG 16.1.2)
			Past Conflict	Fatalities lagged in time
			Political Stability	Political stability and absence of violence/terrorism
			Neighbouring Conflict	Number of neighbouring countries with at least 25 battle-related deaths and conflict status of a country
			Proliferation of Small Arms	Civilian firearms holdings per 100 residents
	Enabling Environment	Resilience	Territorial Integrity	State authority over territory
			Forced Displacement	Refugees + IDPs as a share of population
			Voice and Accountability	Voice and accountability
			Military Capacity	Military expenditure as share of GDP
				Armed forces personnel as share of total labour force

⁵ ESCWA (2023) Arab Risk Monitor: Quantifying the drivers of risk of conflict, version 1.0. Available at <https://www.unescwa.org/publications/arab-risk-monitor-drivers-conflict>.

Pathway	Domain	Dimension	Driver	Variable
Climate Pathway	Natural Resource	Vulnerability	Reliance on Agriculture	Agriculture, forestry and fishing, value added (percentage of GDP)
			Water Availability	Renewable internal freshwater resources per capita (m ³)
		Resilience	Water Stress	Freshwater withdrawal as a percentage of available freshwater resources (SDG 6.4.2)
			Land Stress	Change in share of agricultural land within total land area
	Climate Hazard	Vulnerability	Human Impact of Natural Disasters	Share of population affected by natural disasters (percentage)
				Disaster-induced internal displacement (new displacement)
		Resilience	Adaptation Strategies	Score of adoption and implementation of national Disaster Risk Reduction strategies in line with the Sendai Framework (SDG 13.1.2)
			Adaptation Finance	Climate adaptation-related development finance, as a share of GDP

Pathway	Domain	Dimension	Driver	Variable
Development Pathway	Economy	Vulnerability	Trade Dependence	Dependence on commodity exports
			Financial Dependence	Personal remittances received (percentage of GDP)
				Net official development assistance received (percentage of gross national income)
			Food Security	Cereal import dependency ratio
				Value of food imports in total merchandise exports (percentage) (3-year average)
				Prevalence of undernourishment (percentage) (3-year average) (SDG 2.1.1)
				Prevalence of moderate or severe food insecurity in total population (percentage) (3-year average) (SDG 2.1.2)
			Inequality	Gini index
				Top 10/bottom 50 ratio

Pathway	Domain	Dimension	Driver	Variable
		Resilience	Economic Development	GDP per capita (current US\$)
			Economic Growth	GDP growth (annual percentage)
			Public Debt	General government gross debt (percentage of GDP)
	Society	Vulnerability	Unemployment	Unemployment, total (percentage of total labour force) (modelled ILO estimate)
			Youth Bulge	Share of youth in total adult population
			Infant Mortality	Mortality rate, under 5 (per 1,000 live births)
		Resilience	Education	Mean years of schooling
				Expected years of schooling
			Social Protection	Share of population covered by social protection schemes/systems (SDG 1.3.1)
			Health Coverage	Universal health coverage (service coverage index) (SDG 3.8.1)
			Water and Sanitation Services	People using at least basic drinking water services (percentage of population) (SDG 6.1.1)
				People using at least basic sanitation services (percentage of population) (SDG 6.2.1)
	Institutions	Vulnerability	Corruption	Control of corruption
		Resilience	Rule of Law	Rule of law
			Government Effectiveness	Government effectiveness index

Source: Authors' elaboration.

The Conflict Pathway

The driver for conflict proximity was built following a different approach that considers the presence of conflict in the country itself, rather than focusing solely on whether a country's neighbours were in conflict. The conflict proximity variable is composed of two elements: (a) if the country is in conflict as per the definition presented in the Uppsala Conflict Data Program (more than 25 fatalities in the considered period of one year); and (b) the number of bordering countries that are in conflict (according to the same definition) divided by the total number of bordering countries. The same source of data is used as per the previously calculated driver, yet the newly constructed driver better reflects the reality in parts of the region.

The conflict proximity driver ranges by construction with values between zero and one. Zero means that the country is not in conflict and has no bordering countries in conflict, while one means that the country is in conflict and surrounded by countries in conflict on all its borders. Given the range of values between zero and one, and despite the distribution not being normal, no further normalization was implemented as this categorical driver can be considered as a support indication of conflict vulnerability. In terms of the directionality of the driver with respect to the risk of conflict, a direct correlation is expected as per the previous analysis. A higher value of conflict proximity is directly related to the conflict vulnerability and risk: a country in conflict and surrounded by countries in conflict is more likely to be affected by the risk of conflict continuation and perpetuation in the following years. However, a country that is not in conflict and not surrounded by other active conflicts is unlikely to be vulnerable or at risk of conflict. The formula for calculating the conflict vulnerability dimension remains the same.

The Climate Pathway

Changes have been made in the Climate Pathway both in the natural resource and climate hazards domains. For the natural resource domain, a different normalization approach was used compared to the one presented in the methodological paper. Natural thresholds were not consistent with the other drivers utilized for the risk analysis. Min-max normalization methodology is utilized for the scaling of water availability and the water stress drivers. This adopted methodology represents the reality in terms of natural resources and specifically water scarcity in the Arab region. This change does not affect the formulas utilized for calculating natural resource vulnerability and resilience.

For climate hazards vulnerability, only the variables relating to human impact of climate hazards were included, both in terms of share of population affected and share of displaced population impacted by climate hazard events. The reason not to include the economic impact of climate hazards is mainly related to the scarcity of data for the Arab region and the lack of consistency with the human impact driver. Given the challenges of measuring the economic damages of extreme climate and weather-related events, namely in certain areas, the analysis was narrowed to only include the human impact, which can be considered as a good proxy for the vulnerability of economic, social and institutional systems to climate hazards. The formula for calculating climate hazard vulnerability changed to:

$$(10) \text{ CH Vuln} = (\text{ShAffPop} + \text{Clim IDPs})/2$$

The Development Pathway

For the Development Pathway and in particular economic vulnerability, changes have been implemented with respect to the driver of economic inequality due to a lack of frequent data over time. Two variables were selected to replace the one previously utilized, and they are frequently updated for all Arab countries and at a global level, even if they might discount for a certain level of

imputation, especially for countries lacking data availability. The Gini index was used to measure income inequality across the different components of the population. The second driver for inequality that was utilized is the ratio of income owned by the top 10 per cent divided by the bottom 50 per cent of the population. This driver reflects the level of accumulation of resources and income in the hands of the richer section of the population. The distribution of these two variables is normal, therefore the min-max normalization approach is utilized for both variables. The higher the inequality, the higher the risk and economic vulnerability. The formula for calculating economic vulnerability remains the same.

Within the framework of the economic resilience driver, it was decided not to consider the tax income revenue as a share of the GDP driver as a source of risk, despite its relevance for the analysis, due to the lack of data in the region and the distortion that its inclusion could create in country comparison and representation in this context. Instead, a driver of public debt was used. Data that has been used for this dimension refers to general gross government debt as a share of GDP. Government debt is defined according to the convergence criteria set out in the Maastricht Treaty which comprises currency, bills and short-term bonds; other short-term loans; and other medium- and long-term loans and bonds. This data is retrieved from the International Monetary Fund, and it covers the largest part of countries both at the global and regional levels. This driver can also be considered as a proxy for state capacity where public debt measures the fiscal space to implement public policies and provide public goods and services to the population. It can be considered as a proxy for state capacity where higher debt levels can cause macroeconomic and financial deficits that can reduce a country's capacity to access international markets and institutions. High levels of debt are associated with a higher risk and lower level of economic resilience. The distribution of this driver is close to the normal curve, therefore further normalization is not needed. In terms of directionality, the larger the value of this driver, the lower the resilience in terms of state capacity to cope with the economic and financial needs of the national system. The inverted driver reports on how countries with higher levels of debt have lower economic resilience and therefore face a higher risk of conflict.

Even though no changes have been made to the social vulnerability dimension, several changes were implemented both at driver and variable levels. The only driver that remained for this domain is the social protection driver retrieved from the International Labour Organization dataset, which has data for some countries only after 2015. The driver on education, which was previously considered and retrieved from ESCWA, considers the average between the mean number of years of education and expected years of schooling. These two variables are frequently updated on an annual basis for almost all Arab countries and reflects, in most cases, the changes that happen at the country level even in the case of sudden shocks such as conflict outbreak. There are limitations when using these drivers, especially in certain types of countries, yet the benefits of having these drivers overwhelms the constraints. There are also some limitations related to the path-dependency of these drivers, but their inclusion allows the reporting of annual changes over the years for most of the considered Arab countries. The value of the combined driver is positively related to social resilience in countries with different development challenges and risks, therefore higher levels of these drivers present lower levels of risk. The distribution of these drivers is normal and min-max

scaling methodology was used. The higher the value of this composite driver, the higher the social resilience and lower the risk.

Drivers measuring water and sanitation services were added: people using at least basic drinking water services (percentage of population, SDG 6.1.1) and people using at least basic sanitation services (percentage of population, SDG 6.2.1). These two variables are retrieved from the World Development Drivers, and they are proxies for the level and access to different social services such as water and sanitation infrastructural development, as well as the functioning of service providers that can provide services to the population. The combination of these two variables can be used as a proxy for “water and sanitation services” and combined with the other social resilience drivers: social protection, education and health. The distribution of these variables is normal, therefore the min-max approach can be used for scaling the values from zero to one, where zero means low levels of water and sanitation services, and consequently low levels of social resilience and higher levels of risk.

Health was also included in the social resilience dimension to measure the capacity of public and private entities in a country to cope with the basic needs of the people in this area. The driver that was used is Universal Health Coverage retrieved from the World Health Organization. It can be defined as the average coverage of essential services based on tracer interventions that include reproductive, maternal, newborn and child health, infectious diseases, non-communicable diseases and service capacity and access, among the general and the most disadvantaged population. This driver is measured as an index reported on a unitless scale of 0 to 100, which is computed as the geometric mean of 14 tracer drivers of health service coverage. These drivers are meant to be indicative of service coverage and should not be interpreted as a complete or exhaustive list of the health services or interventions that are required to achieve universal health coverage. The distribution of this driver is normal, therefore the min-max approach was applied to scale the driver from zero to one. This driver is positively correlated with resilience and negatively correlated with risk.

The formula used to include these changes in the drivers is:

$$SocRes = [SocProt + (MeanYearsEdu + ExpYearsEdu)/2 + (WatServ + SanitServ)/2 + HealthCov]/4$$

For the institutional resilience dimension, the source for the governance effectiveness driver was changed from the ESCWA-developed one to the one developed by the World Bank Worldwide Governance Indicators dataset. This change was implemented due to data availability, which allows analysis over time and better reflects the changes that happened, especially as a consequence of shocks. The distribution of this driver is normal; therefore, the min-max methodology was used for scaling this driver. The higher the value of this driver, the higher the institutional resilience and lower the risk. No changes were implemented in the formula for this dimension.



The Arab region grapples with complex and protracted crises aggravated by a multitude of conflicts, climate shocks and development challenges. The damaging effects of these crises impacts people, societies and institutions, and threatens to undo the progress made. This paper presents the key findings of the United Nations Economic and Social Commission for Western Asia Arab Risk Monitor, which assesses vulnerability and resilience to shocks in 22 Arab countries, exploring the issues shaping risks of conflict, crisis and instability. The analysis reveals that weak institutions, combined with the scarcity of natural resources, perpetuate a vicious cycle of instability and conflict. Over the past decade, the region's overall risk has increased, primarily driven by the devastating impact of recent conflict. Water scarcity, food insecurity and corruption emerge as key drivers of high vulnerability while inadequate social protection, water stress and a lack of accountability hamper the region's resilience. Amidst this challenging landscape, efforts to build resilience in the Arab region have shown promise, propelled by the increasing adoption of climate adaptation strategies, the expansion of social safety nets and a significant decline in infant mortality rates. By observing and interpreting structural trends, the Risk Monitor can serve to guide policymakers in the Arab region, development organizations and humanitarian actors who navigate increasingly complex polycrisis dynamics, integrate risk analysis into policymaking and planning, and ultimately advance the Sustainable Development Goals.

